DHAIRYA SHAH

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ **September 2024–May 2026**
Master of Science in Data Science | ***GPA: 3.73/4.0***

K.J. Somaiya College of Engineering | Mumbai, India **July 2020–May 2024**
Bachelor of Technology in Information Technology, with Distinction | ***GPA: 3.62/4.0***

EXPERIENCE

Business Analyst | Street Care, Remote, USA **August 2026–Present**

- Translated outreach team requirements into **6** KPIs for service reach and program activity.
- Analyzed **20K+** outreach records with SQL and Python to find demand trends and coverage gaps.
- Built weekly Excel reports for non-technical leads, cutting ad hoc data requests by **25%**.

Data Operations Analyst | Axion24 AI Technologies, Mumbai, India **May 2025–August 2025**

- Validated field-level data integrity across **~300K** records with Python and SQL, cutting downstream errors by **20%**.
- Processed **~12K** regulatory circulars per month across **3** languages with SQL, Python, and LLM tools, applying data governance rules to catch metadata inconsistencies and duplicate records, cutting manual review volume by **~30%**.
- Built a Power BI analytics layer tracking platform usage and data quality, and defined data quality standards with engineers and stakeholders to sustain a **99%** uptime SLA.

Data Analyst | Accenture, Mumbai, India **January 2024–June 2024**

- Ran scenario analysis and hypothesis testing on promo and pricing changes, delivering recommendations the client adopted that lifted category revenue by **8%** the following quarter.
- Built statistical models in Python, SQL, and SAS for sales reporting and short-term revenue forecasts across product categories and store regions.
- Built Power BI dashboards on category-level sales, promo lift, and regional performance, adopted by the client's merchandising leads to plan weekly markdowns and reorder decisions.

Data Analyst | Expify, Mumbai, India **June 2022–October 2023**

- Designed and led A/B tests on pricing and checkout flows, recommending rollouts that lifted revenue per session by **15%**.
- Designed the Postgres schema and automated weekly reporting across Postgres, HubSpot, Google Analytics, and Excel with Python and SQL, cutting manual prep from **40 to under 8 hours** a week.
- Built **12** Tableau dashboards on customer lifecycle and segmentation, adopted by product and marketing leads to drive weekly business reviews and reduce reporting turnaround by **30%**.

PROJECTS AND RESEARCH

Bank Stress Early Warning System ***Python • SQL • Tableau***

- Built a stress testing framework for 987 US banks on 27 quarters of FFIEC Call Report data, measuring unrealized securities portfolio losses vs capital, uninsured deposits, CRE concentration, and liquidity coverage.
- Backtested the framework against actual failures: all 4 US bank failures over $1B since 2023 ranked in the top 12% before failing.
- Modeled interest rate shock scenarios **(±100bp, +200bp)** on bank securities portfolios to estimate after-tax capital impact as a share of Tier 1.

Options Pricing Lab ***Python • SciPy • Streamlit***

- Built Black-Scholes, binomial tree, and Monte Carlo pricers from scratch and benchmarked **8,568** option contracts across 5 tickers; the American binomial tree cut median pricing error on in-the-money puts by **86% to 88%** vs Black-Scholes.
- Solved implied volatility for every contract and fit SVI curves to build a live 3D volatility surface refreshed daily; near-the-money prices landed within **1%** to **2%** of market midpoint.

Forecasting Macro and Microeconomic Indicators with ML ***Python • LSTM***

- Built Linear Regression, Decision Tree, Random Forest, and LSTM models to forecast India CPI inflation, GDP per capita across **227** countries, and a US equity index; validated with ADF and ACF/PACF stationarity tests.
- Published and presented at Hinweis International Conference ASIT 2024 (**150+** attendees).

SKILLS AND CERTIFICATIONS

Analytics: Data Governance, Data Modeling, Database Design, Statistical & Quantitative Modeling, Hypothesis Testing, A/B Testing, ETL, Monte Carlo Simulation, Financial Analysis, Financial Modeling, Stress Testing, Credit Risk Analysis.
Machine Learning: LSTM, Random Forest, Linear/Logistic Regression, Decision Trees, Time Series Forecasting.
Programming: SQL (Window Functions, CTEs, Joins), Python (Pandas, NumPy, SciPy, Scikit-learn), R, SAS, C++.
Tools: Excel (Pivot Tables, XLOOKUP, Macros), Power BI, Tableau, Looker, Snowflake, Git, AWS, Azure, Databricks, Streamlit, DuckDB, LLM tools (Claude, Gemini).
Certifications: Bloomberg Market Concepts; Databricks Certified Data Analyst Associate; Google Cloud Big Data & ML.